

Pavithra Uthrah R. K.

Computer Science & Engineering Candidate · Data Engineering / Software Engineering

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EXPERIENCE

Data Engineer Intern – KaarTech

Jul 2026 – Present

Chennai, India

- Built an end-to-end PySpark ETL pipeline (explicit schema, validation, cleaning, feature engineering, star schema) for a Power BI-ready FMCG sales analytics dataset, covered by 39 automated pytest tests.
- Defined requirements, source-system model, and Bronze/Silver/Gold Medallion architecture for a Databricks Lakehouse consolidating EV fleet telemetry, charging, and maintenance data across six source systems.
- Designed a proof-of-concept architecture combining Databricks Genie (structured SAP data) with a RAG layer (policy documents) for a natural-language SAP procurement assistant, including a 32-question evaluation set.
- Built a Streamlit prototype using an LLM-based intent classifier to route natural-language requests into data-engineering operations.

Software Development Engineer Intern – AIC-CIIC

May 2026 – Jun 2026

Chennai, India

- Architected the backend of a modular, role-based ERP (Django, PostgreSQL) managing the full startup lifecycle for a startup incubation centre, with RBAC across five roles.
- Built a facility booking backend (Django) serving 140+ incubated startups, with equipment-level conflict detection, approval workflows, and a booking calendar.
- Built an AI-assisted WhatsApp worklog automation pipeline (FastAPI) that extracts structured task data from unstructured messages using an LLM, with a rule-based fallback.

SELECTED PROJECTS

AutCore – 1st Place, HackHub'25 (IEEE CompSoc VIT-C, GitHub & Devfolio)

AI-driven autism screening tool combining computer vision, speech analysis, and symptom profiling. Built the web platform with a teammate in 24 hours after joining the hackathon 12 hours late; placed 1st of 500 teams.

Last-Mile Delivery Tracker (React, TypeScript, Express, PostgreSQL)

Full-stack logistics platform with a configurable pricing engine, zone-based routing, agent assignment, and append-only delivery tracking with historical price snapshots.

Capital Allocation & Risk Optimization Platform (Python, XGBoost, Pyomo)

XGBoost credit-risk model (0.732 ROC-AUC) feeding an integer-programming portfolio optimizer; selected 232 loans and 22 properties, deploying \$4,999,940 of a \$5,000,000 budget.

FMCG Sales Analytics Pipeline (PySpark, Power BI, pytest)

Tested PySpark ETL pipeline transforming raw FMCG retail sales data into a Power BI-ready star schema (1 fact + 6 dimension tables), covered by 39 automated tests.

SKILLS

Languages

Python, SQL, TypeScript, JavaScript, Java, C, C++

Data Engineering

PySpark, Databricks, Delta Lake, Unity Catalog, Medallion Architecture, dbt, Airflow

Data & BI

Power BI, DAX, Tableau, Pandas, NumPy

Machine Learning / AI

XGBoost, RAG, Vector Search, Databricks Genie, LLM Prompt Engineering

Backend & APIs

Django, FastAPI, Express, REST APIs, RBAC, JWT

Databases / Frontend

PostgreSQL, SQLite, React, Tailwind CSS, Bootstrap 5

Cloud & Tools

Azure Databricks, Docker, Git, Alembic, pytest

EDUCATION

Vellore Institute of Technology, Chennai – B.Tech, Computer Science & Engineering

2023 – May 2027

Coursework: Database Management Systems, Data Structures & Algorithms, Operating Systems, Computer Networks, OOP, Software Engineering, Data Analytics

St. Michael's Academy – High School Diploma (ISC / ICSE)

2023

ISC: 80% · ICSE: 92%

LEADERSHIP

Chairperson → Vice Chairperson → Secretary / Design Lead, Open Source Programming Club, VIT Chennai

2024 – Present

Led a 250+ member technical community as Vice Chairperson; helped scale the club to #1 among 110+ student clubs at VIT Chennai, recognized as Best Tech Club 2025-26. Organized 3 national-level hackathons and 10+ workshops; managed a 15-member design team.

Rotaractor, Rotaract Club of VIT Chennai · Student Mentor Volunteer, U&I Trust

CERTIFICATIONS

Google Data Analytics Professional Certificate – Google, via Coursera (January 2026)